

1. Write a function in C that takes an array of integers and its size, and returns the second largest element.

```
#include <stdio.h>

int second_largest(int arr[], int n)
{
    int largest, second_largest;

    if (arr[0] > arr[1])
    {
        largest = arr[0];
        second_largest = arr[1];
    }
    else
    {
        largest = arr[1];
        second_largest = arr[0];
    }

    for (int i = 2; i < n; i++)
    {
        if (arr[i] > largest)
        {
            second_largest = largest;
            largest = arr[i];
        }
        else if (arr[i] > second_largest && arr[i] != largest)
        {
            second_largest = arr[i];
        }
    }

    return second_largest;
}

int main()
{
    int arr[] = {10, 25, 5, 40, 30};
    int n = sizeof(arr) / sizeof(arr[0]);

    printf("Second Largest = %d", second_largest(arr, n));

    return 0;
}
```

Output :
Second Largest = 30

2. You are given an 8-bit register represented as an unsigned char. Write a function to:

```
#include <stdio.h>

unsigned char modifyRegister(unsigned char reg)
{
    reg = reg | (1 << 2);
    reg = reg & ~(1 << 5);
    reg = reg ^ (1 << 0);
    return reg;
}

int main()
{
    unsigned char reg = 0x00;

    reg = modifyRegister(reg);

    printf("Modified register = 0x%02X", reg);

    return 0;
}
```

Output :
Modified register = 0x05

3. Write a C program to print a pyramid of stars for a given number n .

```
#include <stdio.h>

int main()
{
    int num, i, j;
```

```
printf("Enter the number of rows: ");
scanf("%d", &num);

for (i = 1; i <= num; i++)
{
    // Print spaces
    for (j = 1; j <= num - i; j++)
    {
        printf(" ");
    }

    // Print stars
    for (j = 1; j <= (2 * i - 1); j++)
    {
        printf("*");
    }

    printf("\n");
}

return 0;
}
```

Output :

Enter the number of rows: 5

```
*
***
*****
*****
*****
```